How Criminals Make Decisions: A Microeconomic Perspective

EC-224 · Intermediate Microeconomics

Prof. Naveen Sunder

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About Me

Naveen Sunder · Bentley University

- Grew up in New Delhi, India.
- Masters and PhD in Economics, Cornell University; subsequently worked at Harvard. Three years of industry experience before graduate school.
- Research: economic and statistical tools applied to health and education outcomes in Africa and Asia.
- Outside work: soccer, TV, podcasts. Six languages; learning a seventh.

My research asks: *how do people make decisions under uncertainty?* Today's lecture uses the same framework — applied to criminals, firms, and AI companies.

Two suspects. Separate rooms. Same decision.

Applying the same logic to a different context. . .

Think about this:

Two tech companies are caught using each other's stolen AI training data. Interrogated *separately* by regulators.

Do you confess — or stay silent?

- **Both silent:** moderate fines for each.
- **One confesses, one silent:** confessor gets immunity; the other is crushed.
- **Both confess:** both penalised heavily.

Hold that thought. We will build the formal model for it.

Game Theory

Game theory models **strategic behaviour** by agents who understand that their actions affect others' actions, and hence the outcome of the game.

A **game** consists of:

- A set of **players**
- A set of **strategies** available to each player
- A **payoff function** giving each player's outcome for every combination of strategies

Key idea: Your best choice *depends on what others do.* That interdependence is what makes this game theory, not just individual optimisation.

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Where Does Today Fit?

Consumer Choice → Firm Behaviour → Market Structure → **Game Theory** ⋆ You are here → Information + Uncertainty

The Payoff Matrix | Representing our situation in numbers...

	Clyde: Silent (S)	**Clyde: Confess (C)**
Bonnie: Silent (S)	$-5,\ -5$	$-30,\ -1$
Bonnie: Confess (C)	$-1,\ -30$	$-10,\ -10$

- Numbers = years in prison (negative = bad).
- Convention: *(Bonnie's payoff, Clyde's payoff)*.
- **Same matrix applies to the AI example:** replace "confess" with "disclose to regulators." The mathematics is identical.

"Solving" the Game: Step 1 | What is Bonnie's **best response**?

Step 1: Fix Clyde's strategy — what does Bonnie do?

For each of Clyde's strategies, find Bonnie's best payoff.

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- If Clyde plays **Silent**: $-1 > -5 \Rightarrow$ **Bonnie's BR: Confess**
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Confess is Bonnie's best response regardless of what Clyde does. This is called a **dominant strategy**.

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"Solving" the Game: Step 2 | What is Clyde's **best response**?

Step 2: Fix Bonnie's strategy — what does Clyde do?

Now look at Clyde's payoffs (second number) in each row.

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"Solving" the Game: Step 3 | Combine best responses ⇒ **Nash Equilibrium**

Step 3: Where do both best responses coincide?

The Nash Equilibrium is the cell where *both* players are playing their best response simultaneously.

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Nash Equilibrium: (Confess, Confess) $= (-10,\ -10)$.

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Practice Question

Find the Nash Equilibrium:

Players A and B. A chooses Top or Bottom; B chooses Left or Right. Underline the best payoff for each player in each column (for A) and each row (for B). Where do the underlines meet?

	Player B: Left	**Player B: Right**
Player A: Top	3, 9	2, 8
Player A: Bottom	0, 2	1, 1

Hint: Top dominates Bottom for A. Left dominates Right for B.

Simultaneous vs. Sequential Games

Simultaneous

Both players choose **at the same time**, without observing the other's action.

Example: Bonnie & Clyde in separate rooms.

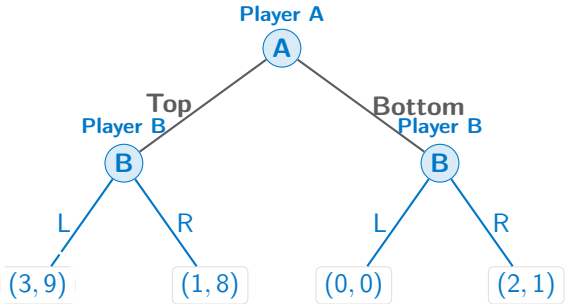
Sequential

One player moves **first**; the other observes and then responds.

Example: Apple launches; Google responds.

Sequential Games: How Do We Solve This?

A moves first. B observes, then responds.



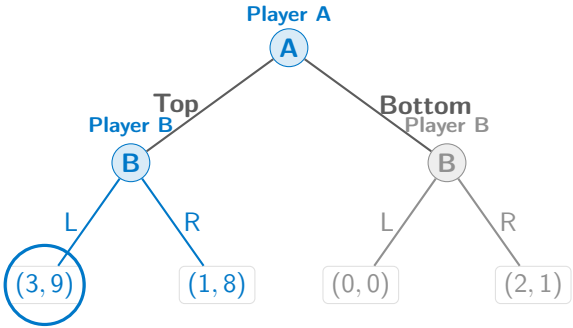
The game tree shows every possible sequence of moves and the resulting payoffs (Player A, Player B).

Think about it:

How would you find the **best strategies** for each player? Where would you start?

Backward Induction: Step 1

Start at the end — what does B do?



Step 1a

Suppose A played Top.

B now chooses Left or Right.

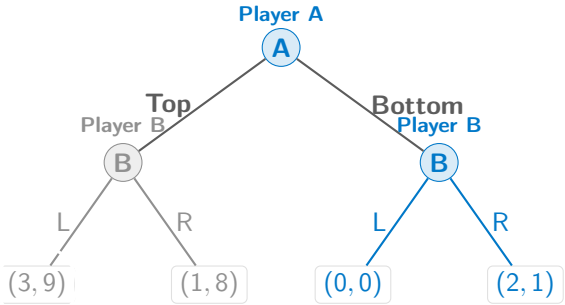
Left: B gets **9**
Right: B gets 8

$\Rightarrow$ **B picks Left.**

Outcome if A plays Top:
(3, 9) $\leftarrow$ B's best response

Backward Induction: Step 1 (continued)

Now suppose A played Bottom instead.



Step 1b

Suppose A played Bottom.

B now chooses Left or Right.

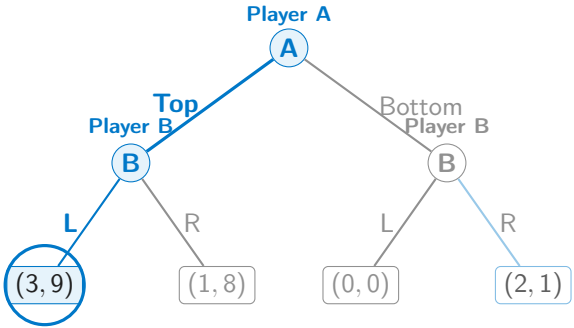
Left:	B gets 0
Right:	B gets **1**

$\Rightarrow$ **B picks Right.**

Outcome if A plays Bottom:
(2, 1) $\leftarrow$ B's best response

Backward Induction: Step 2

Now work back to A — what does A choose?



Step 2

A knows B's responses in advance.

A's real choice:

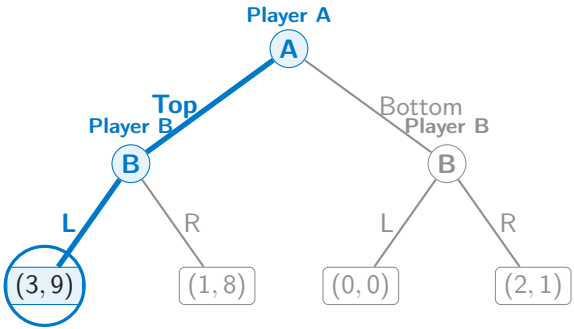
Top	$\Rightarrow$ A gets **3**
Bottom	$\Rightarrow$ A gets 2

$\Rightarrow$ **A plays Top.**

A *anticipates* B's response before moving. This is the logic of **backward induction.**

Backward Induction: The Equilibrium

A plays Top, B plays Left.



Equilibrium Path:

A plays **Top**
B plays **Left**
Outcome: **(3, 9)**

Neither player wants to deviate:

A: switching to Bottom gives $2 < 3$
B: switching to Right gives $8 < 9$

First-mover advantage: by committing to Top, A induces the favourable response from B. This is the **Stackelberg** intuition.

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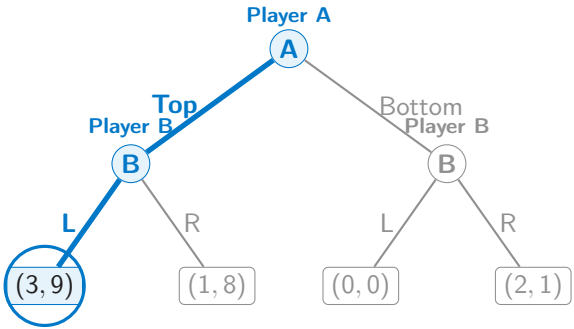

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Game Theory Is Everywhere

Business & Technology

- AI development races
  OpenAI vs. Google vs. Meta
- App store pricing battles
  Apple vs. Spotify
- Salary negotiations
- Real estate bidding wars

Policy & Beyond

- Antitrust & regulation
- Climate agreements
  Countries as players
- Electoral strategy
- International trade policy

Did you know? Game theory was used to design the US radio spectrum auction — a market worth over $60 billion. Not abstract theory: applied in real policy and business decisions every day.

A Critical Assumption: Rationality

Are players really rational in the real world?

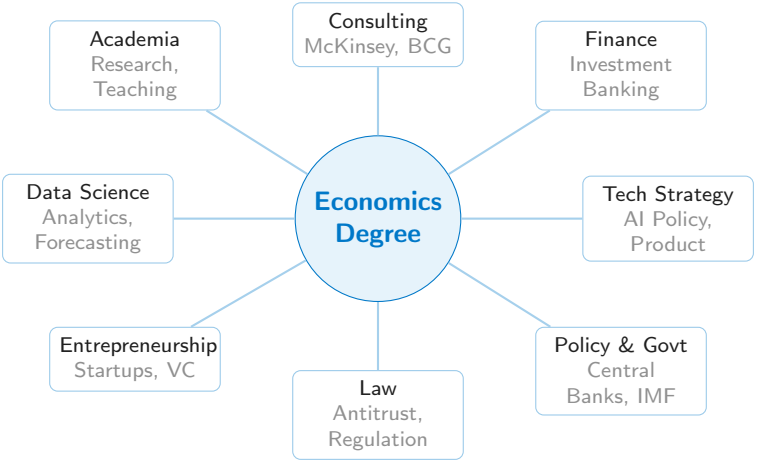
Nash Equilibrium assumes players are **rational**: they always choose the action that **maximises their own payoff**, given what others do.

Think on your own...

Can you think of a real decision — by a person, a firm, or a government — that seemed **irrational**? What was the cost?

Behavioural economics relaxes the rationality assumption — one of the fastest-growing areas of economics, directly relevant to marketing, finance, and public policy.

Economics Is Everywhere



Thank You

Questions?

Name: Naveen Sunder

Email: nsunder@bentley.edu

Website: www.naveensunder.com

Hope to see you in EC-224 in a future semester.

Scan for resources

naveensunder.github.io/IntMicroMiniLecture